CHNT31 · Digital Creative Award · Poster

Code is Data, Too

From open-archaeo via Wikidata to chublets.software: a FAIR software marketplace for cultural heritage

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1 · Software hides in plain sight

Tools for archaeology and heritage research sit in lists, repositories and papers. Their metadata are thin and uneven, so the software is hard to find, cite and link with research data.

The idea: chublets

A **chublet** is a small, documented nugget of research software, described as Linked Data in one shared schema. **chublets.software** is the Cultural Heritage Hub for chublets: a FAIR software marketplace for cultural heritage.

The life of one chublet: from a list to a marketplace

1. **open-archaeo** — 416 software entries
2. **Prepare** — OpenRefine and Python, one vocabulary
3. **Wikidata** — chublet class, WikiProject, seed items
4. **Static demo** — wikibase-static: pages, RDF, SPARQL
5. **Federation Wikibase** — chublets-wikibase.de, prototype
6. **chublets.software** — the hub: a FAIR software marketplace

SPARQLing Unicorn QGIS Plugin

Q360 · item of the chublets Wikibase · 39 statements

instance of	software	1 reference
	utility software	1 reference
	QGIS plugin	1 reference
	Little Minion	1 reference
	chublet	1 reference
Wikidata equivalent	Q74005133	1 reference
developer	Florian Thiery	1 reference
	Timo Homburg	1 reference
	Research Squirrel Engineers Network	1 reference
programmed in	Python	1 reference
source code repository URL	https://github.com/sparqlunicorn/sparqlunicornGoesGIS version control system: Git · web interface software: GitHub	1 reference
software version identifier	0.17.4 preferred rank publication date: 2025-12-19 · version type: stable version and 24 earlier versions, each with its own date and references	2 references
DOI	10.5281/ZENODO.13828632	1 reference
official website	https://plugins.qgis.org/plugins/sparqlunicorn/	1 reference

and 1 more kind of statement: title

class · item = value is a class or an item; plain text = literal value; grey line = qualifier of the statement above



Wikidata properties

software properties of the WikiProject and further properties used for software

114 properties with an equivalent in Wikidata (P4)

CodeMeta terms

the CodeMeta vocabulary for software metadata mapped to Wikidata in the crosswalk

83 terms, 37 with a Wikidata property

Wikibase of chublets: 168 properties

all of them local properties; each states its equivalents (Wikidata P4, CodeMeta P101, schema.org P102)

37	46	52	18	15
crosswalked	CodeMeta only	Wikidata only	chublets addition	built-in and later added

The model is not invented from scratch: it starts from the crosswalk between the two vocabularies.

Sources of the modelling

- **CodeMeta–Wikidata crosswalk** — published, read in full
  68 CodeMeta terms reviewed against Wikidata and further schemas
- **CodeMeta terms page** — published, read in full
  15 further terms that have no row in the crosswalk
- **Wikidata WikiProject Informatics, software properties** — transcribed by hand
  63 Wikidata properties for software (the registry identifiers are not all listed); after review 9 properties were added to the model
- **Live Wikidata check** — checked against Wikidata
  label and datatype of 137 Wikidata entities: 246 checks, 14 differences
- **Decision tables** — decided by hand
  domain and range of 149 properties, schema.org term of 84 properties
- **open-archaeo list** — published, mapped by rule
  416 software entries, with their people, tags and works, mapped to items and statements

What they feed

- **168 properties** with their metadata
- **13 classes** built in and defined
- **826 items** with statements

The model as CSV tables

1007 entities: 13 classes, 168 properties, 826 items
6873 statements - repository chublets-wikibase-demo

Static
wikibase-static - read-only HTML pages

- **What it does**: reads the CSV tables and renders the model as web pages
- **Needs**: no server, no account and no write access to any Wikibase
- **Wikidata items**: shown as they are in the model, including stubs
- **Result**: pages that run on GitHub Pages

Federation
wikibase-federation - writes into a real Wikibase

- **What it does**: pushes the model into a Wikibase in three phases
- **Needs**: an account of the target Wikibase; a dry run is the default
- **Wikidata items**: become federated values; stubs are not created
- **Result**: a live Wikibase; the new ids are recorded in an id map

1. **Ingest** — Wikidata items, FDOx objects and Git repositories
2. **Model** — 168 properties from Wikidata and CodeMeta
3. **Curate & Link** — 826 items and 6873 statements, curated and linked
4. **Export & Publish** — CodeMeta, DCAT and DataCite for the software marketplaces

People


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Ronny Gey


Leif Lobinsky


Lutz K. Schubert

Explore

Browse the chublets

chublets Wikibase

Behind the work

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References

- Thiery, F. *wikibase-static*, version 0.19.0. n4o-rse/wikibase-static
- *chublets-wikibase-demo*. n4o-rse/chublets-wikibase-demo
- *open-archaeo*. n4o-rse/open-archaeo
- ConnOSS crosswalks to CodeMeta, v1.2.0, CC BY 4.0. doi:10.5281/zenodo.21935688
- *find.software*. find-software.org